R109/A_tied_late | Locked full18 results

Same unique VAL checkpoint on VAL/TEST/LIB; native/FP64 ind; historical NSS weights disclosed.

epoch 26 | update 37440 | seed2 | fresh initialization

Data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
VAL ALL	71.1025	66.2037	11.416634	11.427978	11.371692	-0.1928	+0.3948	0.4047
TEST ALL	71.3852	66.1802	11.373796	11.385299	11.329932	-0.1979	+0.3852	0.3940

Dataset	n	NSS Top1 (%)	R109 Top1 (%)	NSS cost	R109 cost
TSP TEST	4500	45.888889	46.911111	7.970400719	7.971084261
CVRP TEST	4500	45.977778	48.311111	18.004802368	18.009786601
TSPLIB	49	20.408163	22.448980	8.217907074	8.215798852
CVRPLIB	100	70.000000	61.000000	67.101111717	67.190010729

NSS: R79 weights trained on historical70114 TRAIN, selected using historical VAL; not retrained on new90k.

NSS TEST predictions were freshly computed on these4500 instances; old3895 TEST metrics are not reused.

SBS is fitted on the new81000 TRAIN only. Oracle is the per-instance legal-pool minimum.

ALL is the equal-problem mean; mixed raw cost is descriptive. TEST informed prior research.

Method selection and true winner use separate colors on the same axes; Top1 is per-instance accuracy.

Selection head: original_score_mlp_maximin | objective: original R77 native CE

R109/A_tied_late | VAL: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.1025	66.2037	11.416634	11.427978	11.371692	-0.1928	+0.3948	0.4047	per-problem TRAIN SBS
TSP	46.5111	33.1111	8.024735	8.045354	7.990533	-0.2563	+0.4280	0.4184	T2T
CVRP	46.6667	18.7111	18.160246	18.269377	18.064087	-0.5973	+0.5323	0.7091	RELD_CVRP
ATSP	76.3333	47.4889	1.638847	1.676905	1.632471	-2.2695	+0.3906	0.3860	UNICO_MatPOENet
OVRP	41.3778	35.7556	7.461405	7.469579	7.415777	-0.1094	+0.6153	0.6300	MoSES_RF
VRPB	81.2889	80.8444	9.355560	9.354876	9.318750	+0.0073	+0.3950	0.3961	reld_mt
VRPL	37.0667	33.0889	12.432069	12.437959	12.383444	-0.0474	+0.3927	0.4003	MoSES_RF
VRPTW	80.1111	79.9556	18.818903	18.819431	18.766139	-0.0028	+0.2812	0.2971	reld_mt
OVRPTW	83.8889	83.7778	10.143229	10.143730	10.116859	-0.0049	+0.2607	0.2575	reld_mt
OVRPB	88.4444	88.5111	6.358814	6.357254	6.341078	+0.0245	+0.2797	0.2915	reld_mt
OVRPL	42.2667	35.9111	7.436680	7.445709	7.390416	-0.1213	+0.6260	0.6421	MoSES_RF
VRPBL	78.8222	78.5111	9.369740	9.371523	9.328366	-0.0190	+0.4435	0.4479	reld_mt
VRPBTW	78.7333	78.4222	19.944790	19.950430	19.851084	-0.0283	+0.4720	0.4524	reld_mt
VRPLTW	78.9333	78.7556	18.763945	18.765871	18.707843	-0.0103	+0.2999	0.3160	reld_mt
OVRPBL	89.3556	89.2000	6.369170	6.370093	6.352824	-0.0145	+0.2573	0.2677	reld_mt
OVRPBTW	83.5778	83.4444	10.660322	10.661098	10.623245	-0.0073	+0.3490	0.3222	reld_mt
OVRPLTW	83.5333	83.4889	10.157089	10.157423	10.129818	-0.0033	+0.2692	0.2719	reld_mt
VRPBLTW	79.7333	79.4000	19.792892	19.797044	19.706420	-0.0210	+0.4388	0.4262	reld_mt
OVRPBLTW	83.2000	83.2889	10.610969	10.609955	10.571302	+0.0095	+0.3752	0.3524	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R109/A_tied_late | TEST: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.3852	66.1802	11.373796	11.385299	11.329932	-0.1979	+0.3852	0.3940	per-problem TRAIN SBS
TSP	46.9111	34.5333	7.971084	7.989621	7.937593	-0.2320	+0.4219	0.4135	T2T
CVRP	48.3111	17.9333	18.009787	18.124983	17.918509	-0.6356	+0.5094	0.6674	RELD_CVRP
ATSP	76.6667	47.2444	1.635380	1.674207	1.629441	-2.3192	+0.3644	0.3633	UNICO_MatPOENet
OVRP	40.6889	33.6444	7.446093	7.455750	7.400487	-0.1295	+0.6162	0.6254	MoSES_RF
VRPB	80.5556	80.4000	9.350246	9.351810	9.315298	-0.0167	+0.3752	0.3813	reld_mt
VRPL	38.1111	32.2667	12.348816	12.356138	12.300958	-0.0593	+0.3891	0.3956	MoSES_RF
VRPTW	81.1333	80.8444	18.689137	18.690545	18.637408	-0.0075	+0.2776	0.2891	reld_mt
OVRPTW	83.8889	83.9333	10.092884	10.093063	10.068135	-0.0018	+0.2458	0.2365	reld_mt
OVRPB	88.0889	88.1333	6.370525	6.370826	6.352225	-0.0047	+0.2881	0.2975	reld_mt
OVRPL	42.1556	34.6000	7.462991	7.472542	7.418033	-0.1278	+0.6061	0.6161	MoSES_RF
VRPBL	79.4000	79.4222	9.396882	9.396379	9.355967	+0.0054	+0.4373	0.4404	reld_mt
VRPBTW	78.8222	78.7556	19.665359	19.665725	19.577666	-0.0019	+0.4479	0.4470	reld_mt
VRPLTW	79.4222	79.2444	18.803171	18.803136	18.745143	+0.0002	+0.3096	0.3280	reld_mt
OVRPBL	89.2667	89.2889	6.375667	6.375741	6.359310	-0.0012	+0.2572	0.2680	reld_mt
OVRPBTW	84.8889	84.8222	10.659658	10.659591	10.624748	+0.0006	+0.3286	0.3018	reld_mt
OVRPLTW	82.9778	82.8667	10.085106	10.086112	10.059001	-0.0100	+0.2595	0.2591	reld_mt
VRPBLTW	79.9111	79.7333	19.704127	19.707068	19.613284	-0.0149	+0.4632	0.4472	reld_mt
OVRPBLTW	83.7333	83.5778	10.661412	10.662149	10.625574	-0.0069	+0.3373	0.3155	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R109/A_tied_late | LIB and locked provenance

Current R109 factory, same buffers and selected checkpoint; no solver execution or LIB tuning.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
TSPLIB	22.4490	10.2041	8.215799	8.240435	8.142167	-0.2990	+0.9043	0.9449
CVRPLIB	61.0000	45.0000	67.190011	67.245364	66.945093	-0.0823	+0.3658	0.5612

SHA256: 2ba5345464b9b022974cc6bf91303a4728376e3b91b59948921bee8122127b96

Unique selection: strict minimum whole18 VAL macro_vs_sbs_pct; first exact tie; common full-TRAIN SBS

Inference: original score MLP utility differences and maximin; no calibration or solver quotas

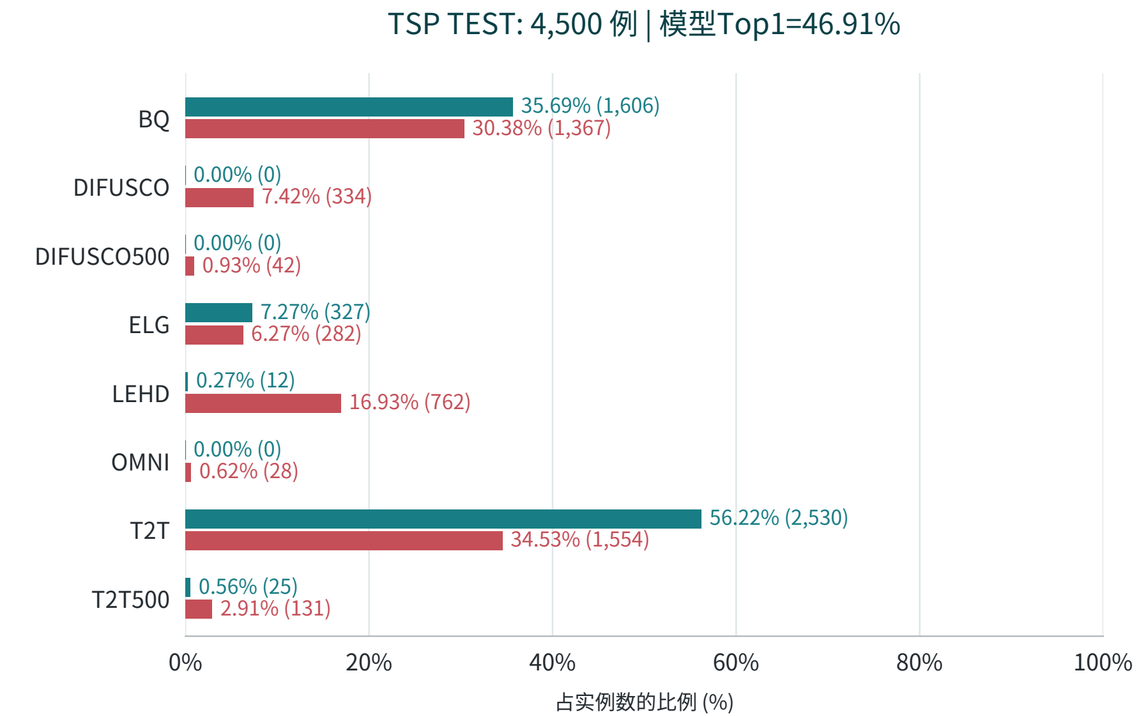
Normalization: new81000 TRAIN only; no TEST/LIB refit.

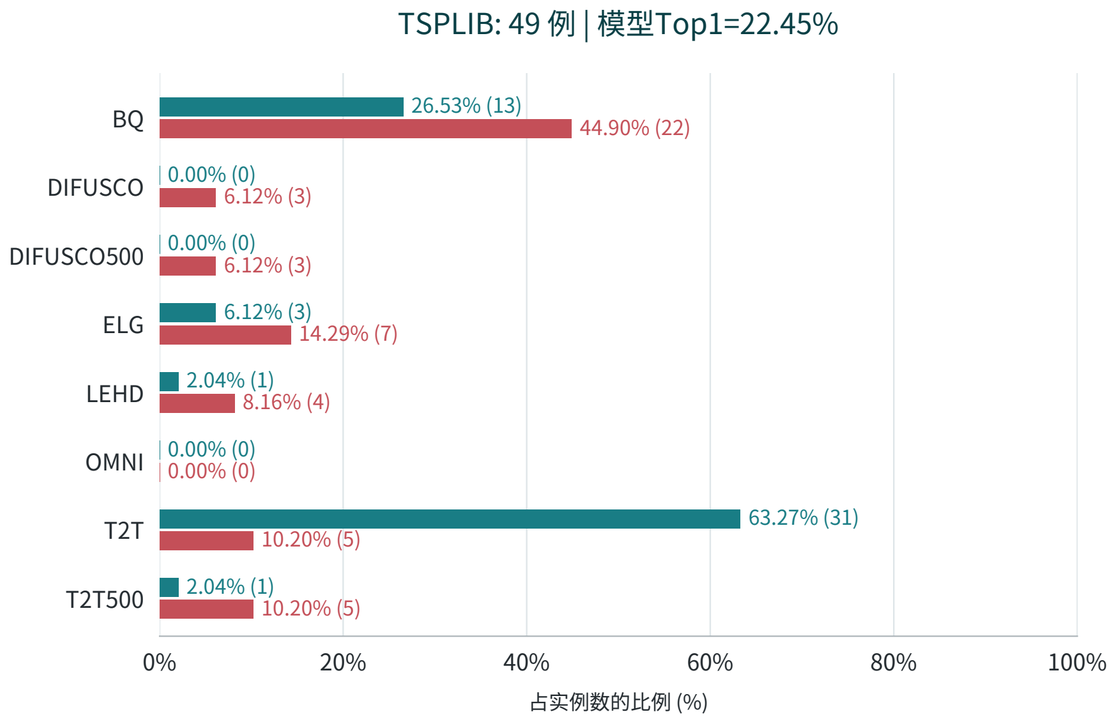
R109/A_tied_late | TSP | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 2ba5345464b9b022

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/TSPLIB | native/FP64 ind





标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

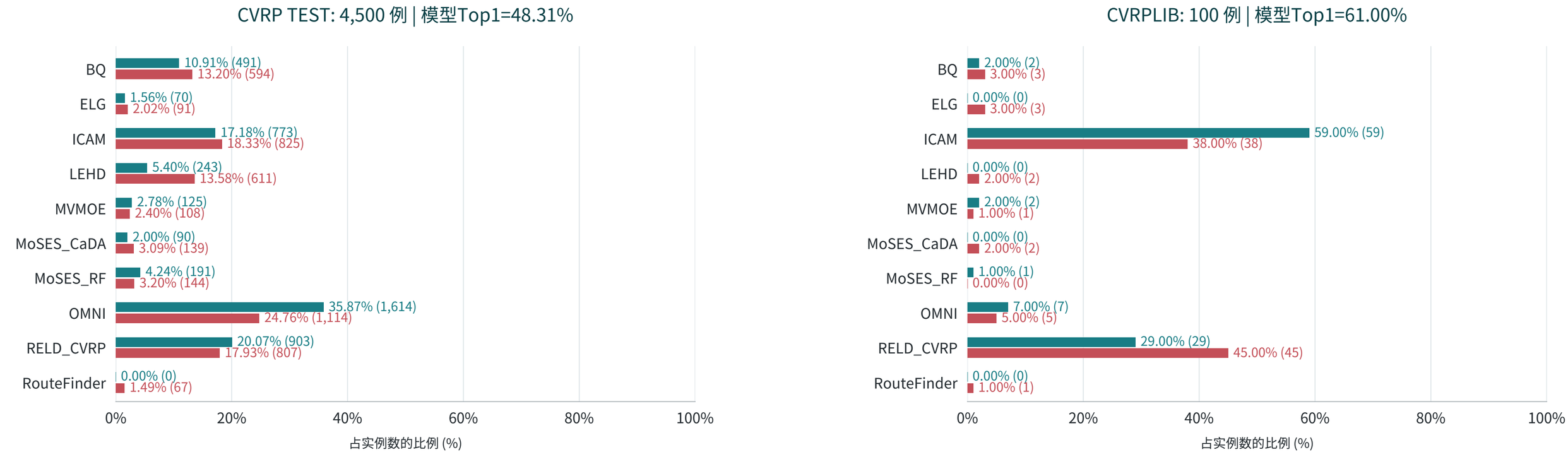
R109/A_tied_late: stable global solver-ID score ties. Same R109 lock; LIB inference only; no training/reselection/Rxx mixing.

R109/A_tied_late | CVRP | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 2ba5345464b9b022

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/CVRPLIB | native/FP64 ind



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

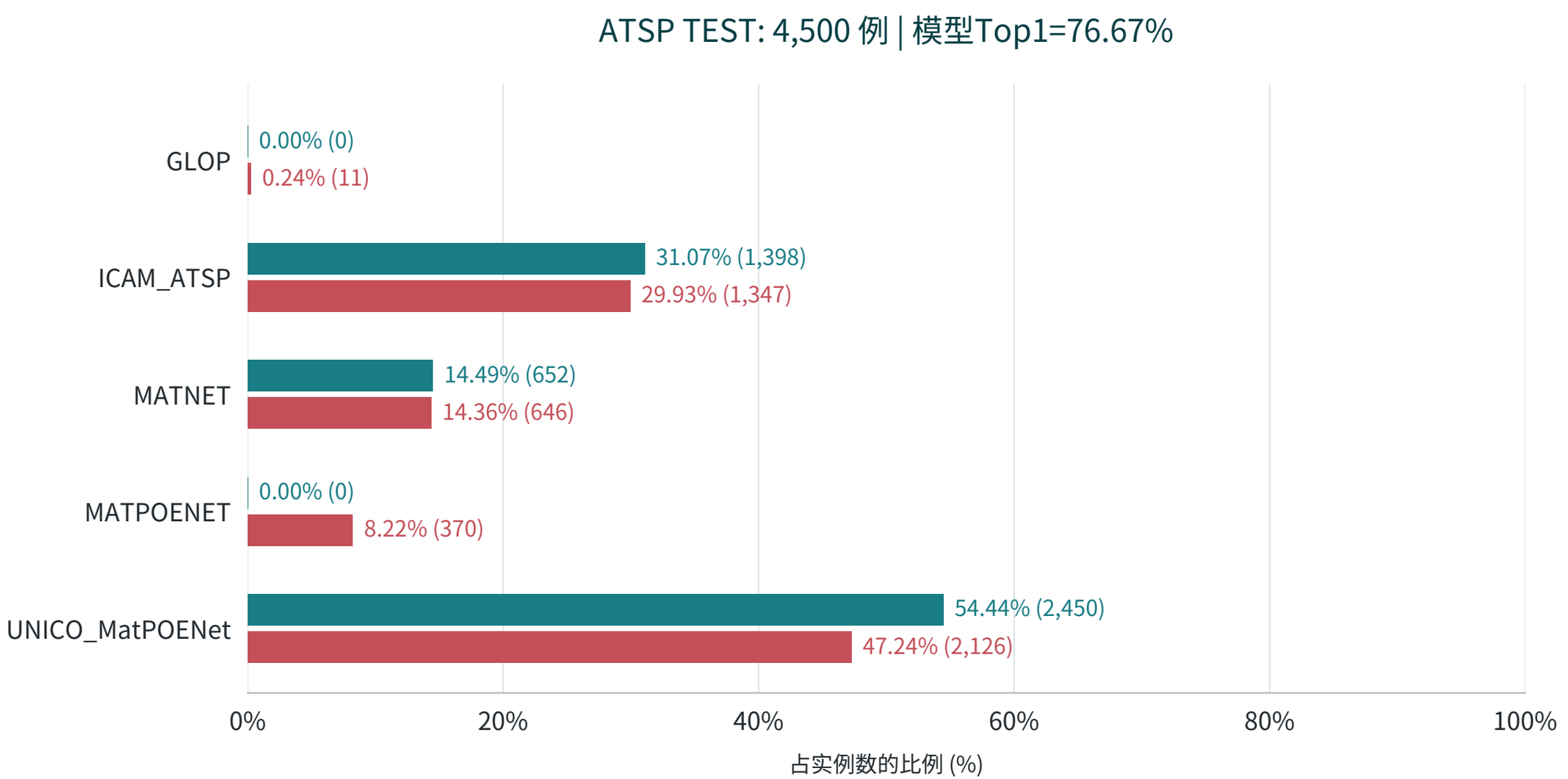
R109/A_tied_late: stable global solver-ID score ties. Same R109 lock; LIB inference only; no training/reselection/Rxx mixing.

R109/A_tied_late | ATSP | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 2ba5345464b9b022

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

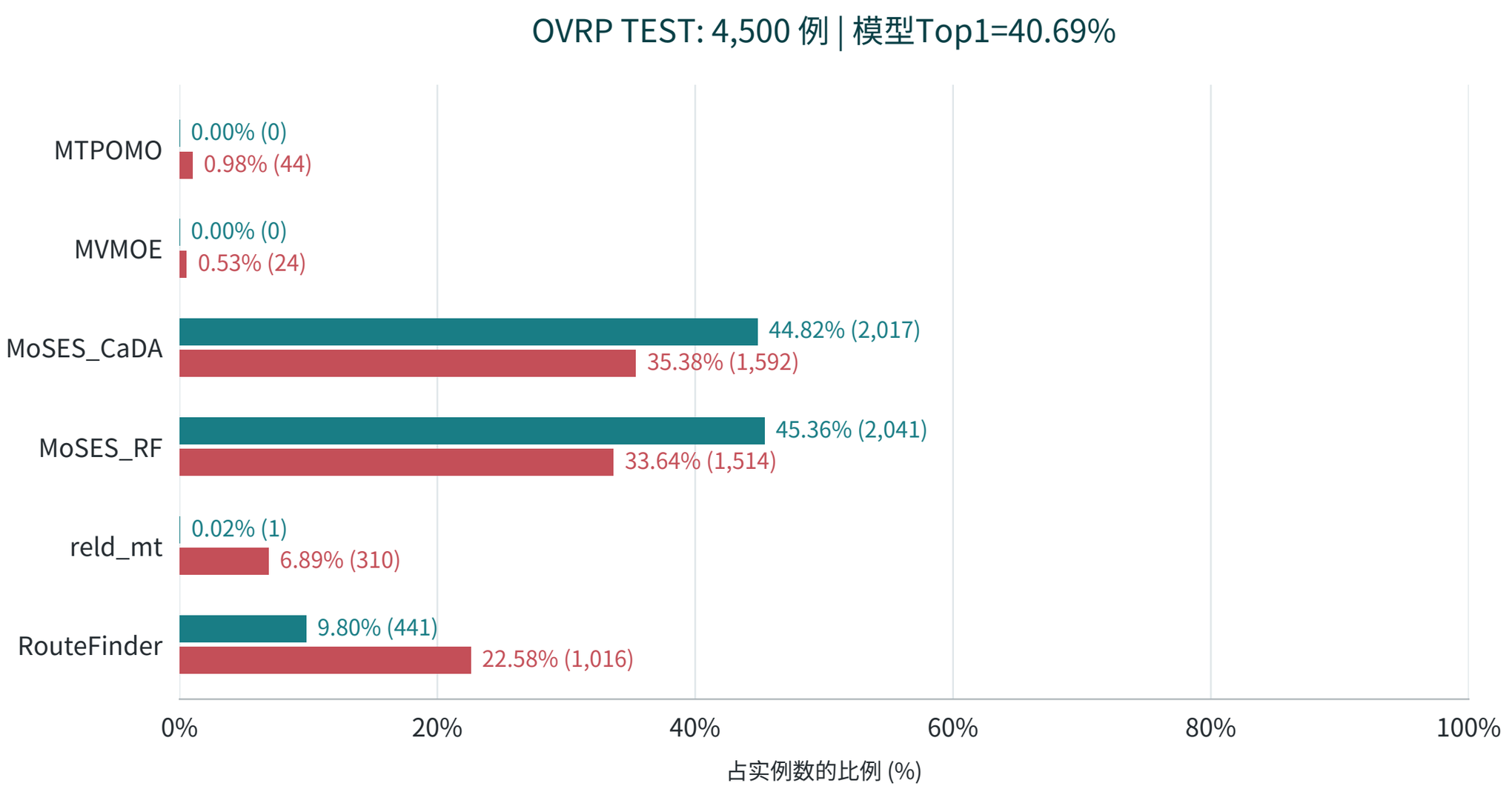
R109/A_tied_late: stable global solver-ID score ties. Same R109 lock; LIB inference only; no training/reselection/Rxx mixing.

R109/A_tied_late | OVRP | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 2ba5345464b9b022

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

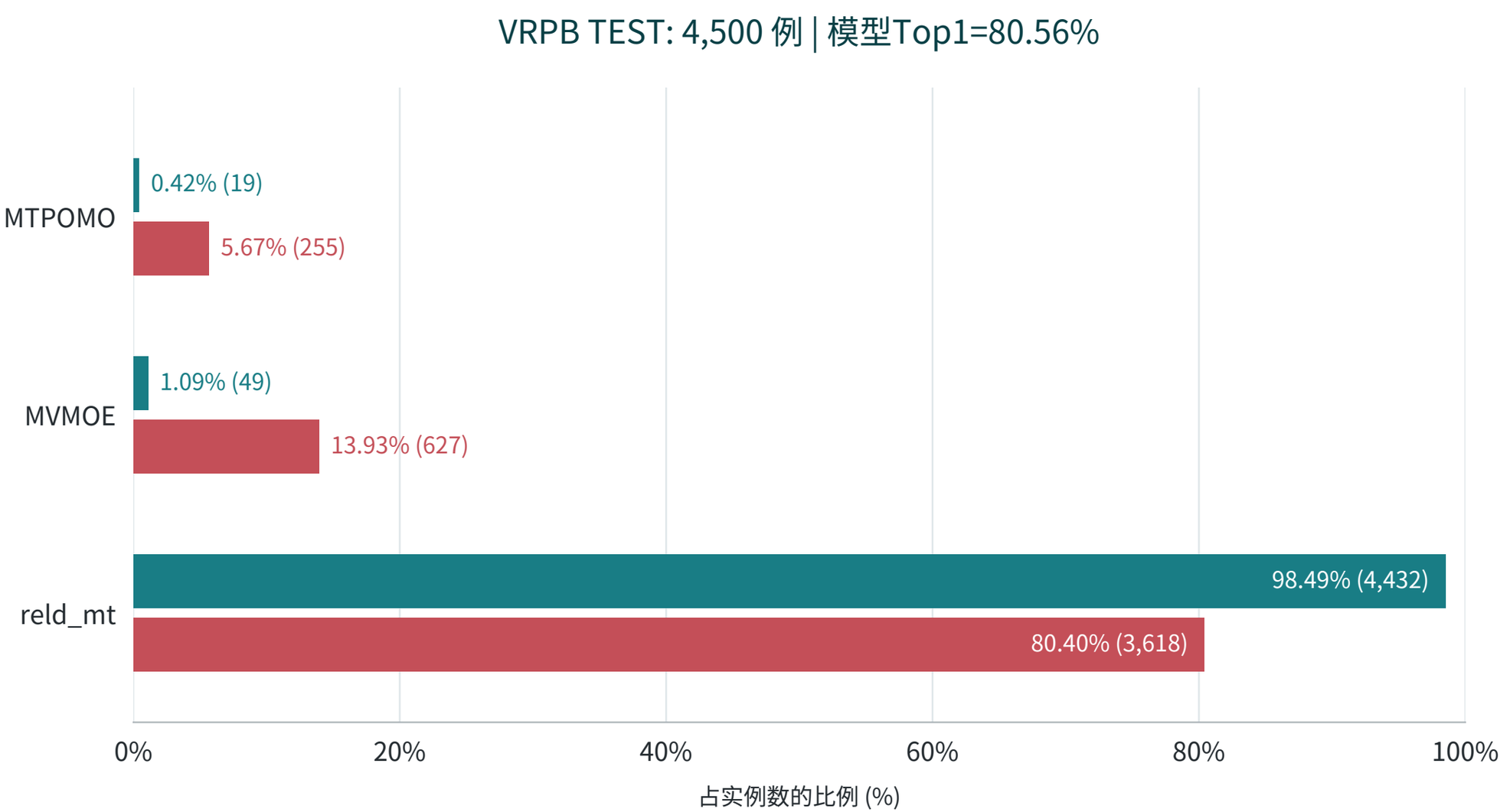
R109/A_tied_late: stable global solver-ID score ties. Same R109 lock; LIB inference only; no training/reselection/Rxx mixing.

R109/A_tied_late | VRPB | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 2ba5345464b9b022

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

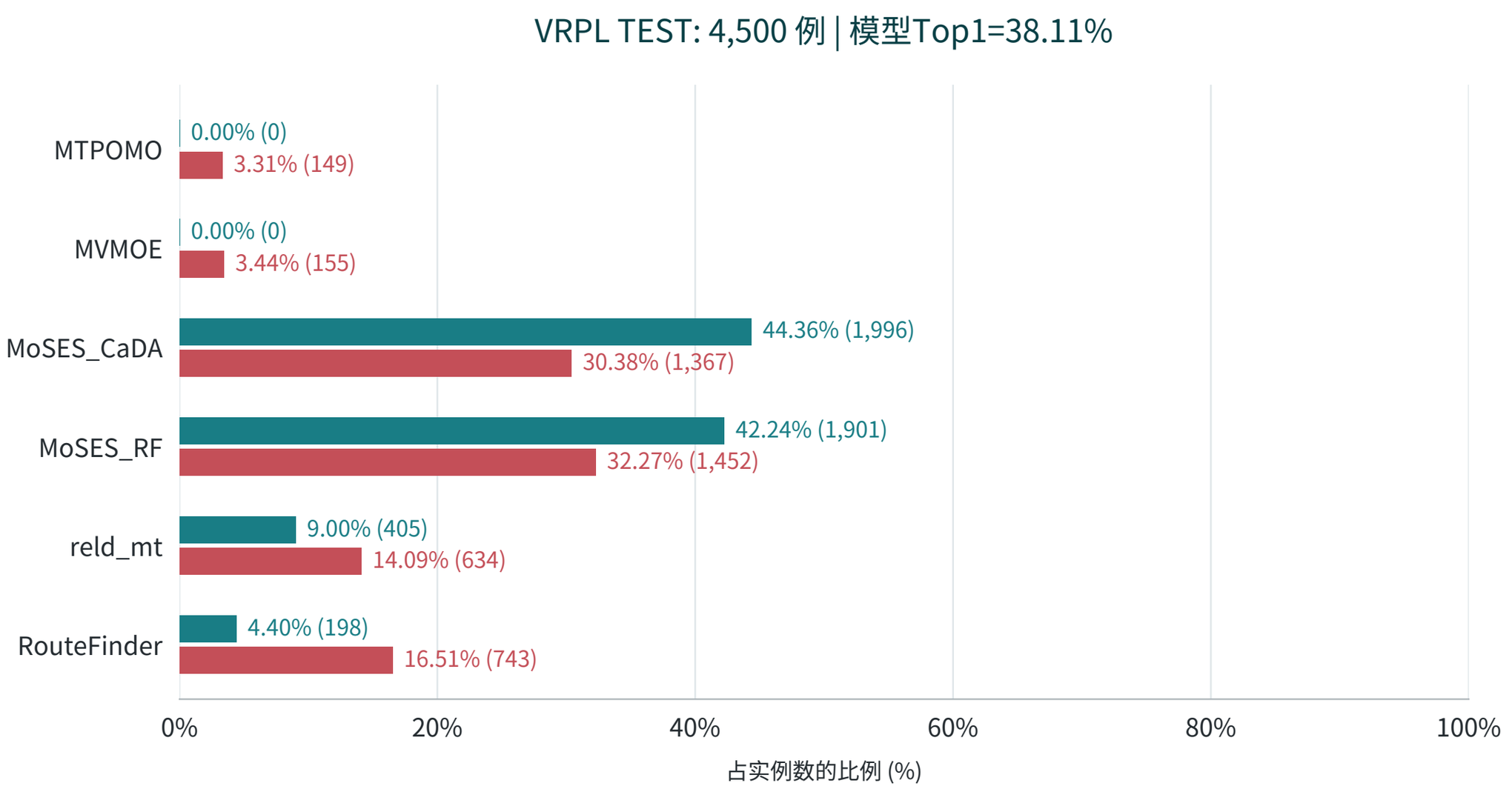
R109/A_tied_late: stable global solver-ID score ties. Same R109 lock; LIB inference only; no training/reselection/Rxx mixing.

R109/A_tied_late | VRPL | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 2ba5345464b9b022

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

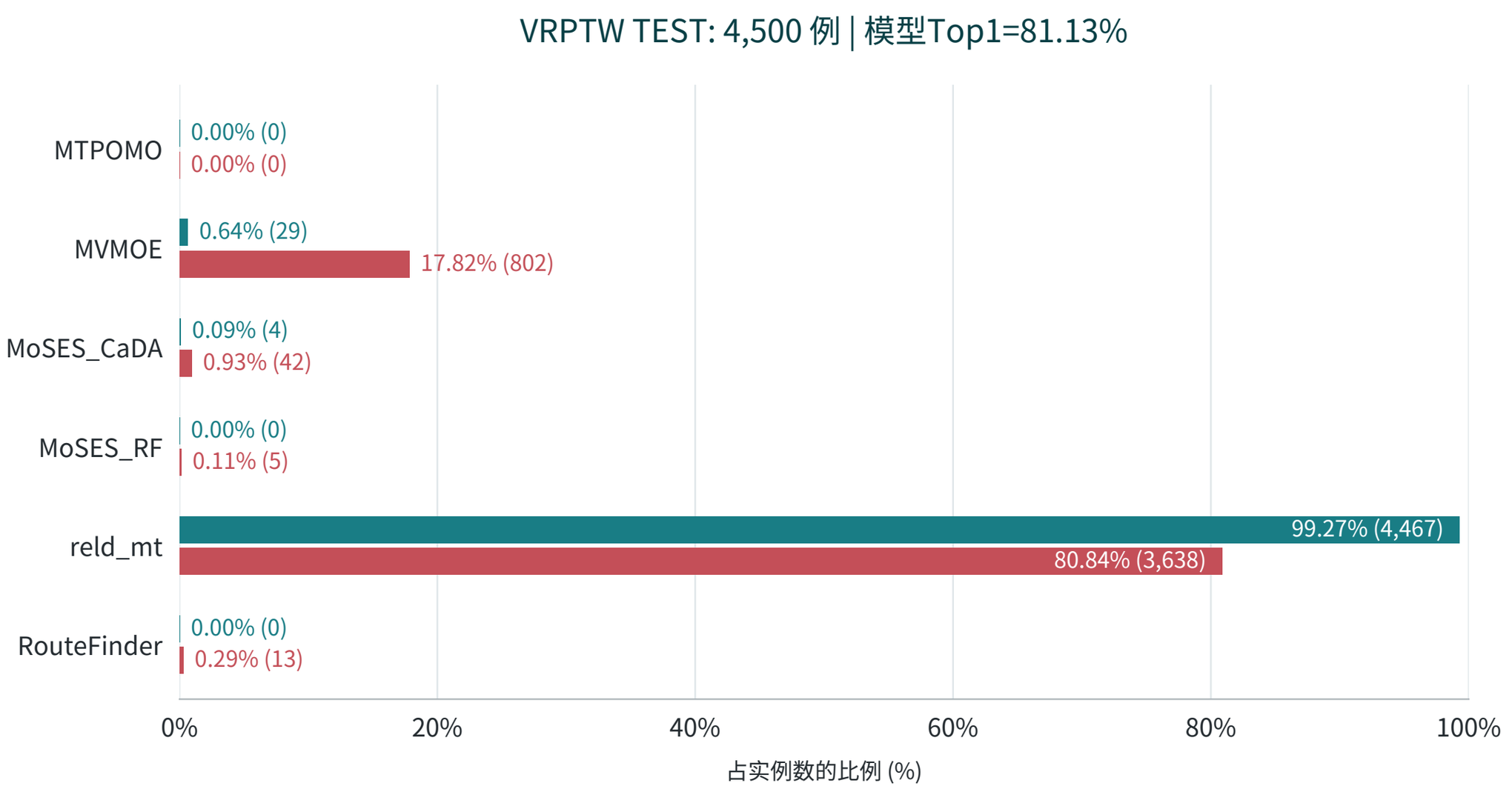
R109/A_tied_late: stable global solver-ID score ties. Same R109 lock; LIB inference only; no training/reselection/Rxx mixing.

R109/A_tied_late | VRPTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 2ba5345464b9b022

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

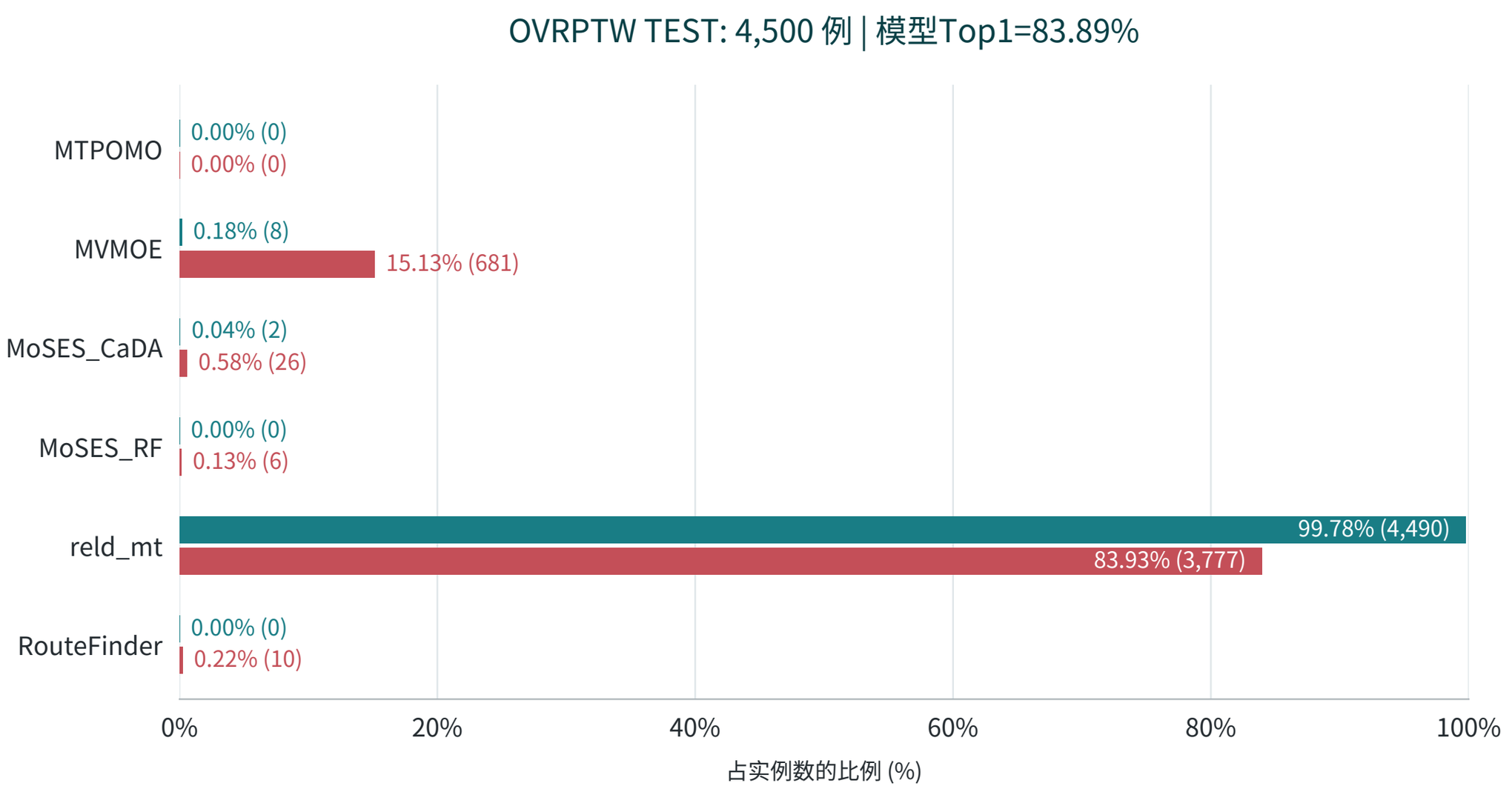
R109/A_tied_late: stable global solver-ID score ties. Same R109 lock; LIB inference only; no training/reselection/Rxx mixing.

R109/A_tied_late | OVRPTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 2ba5345464b9b022

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

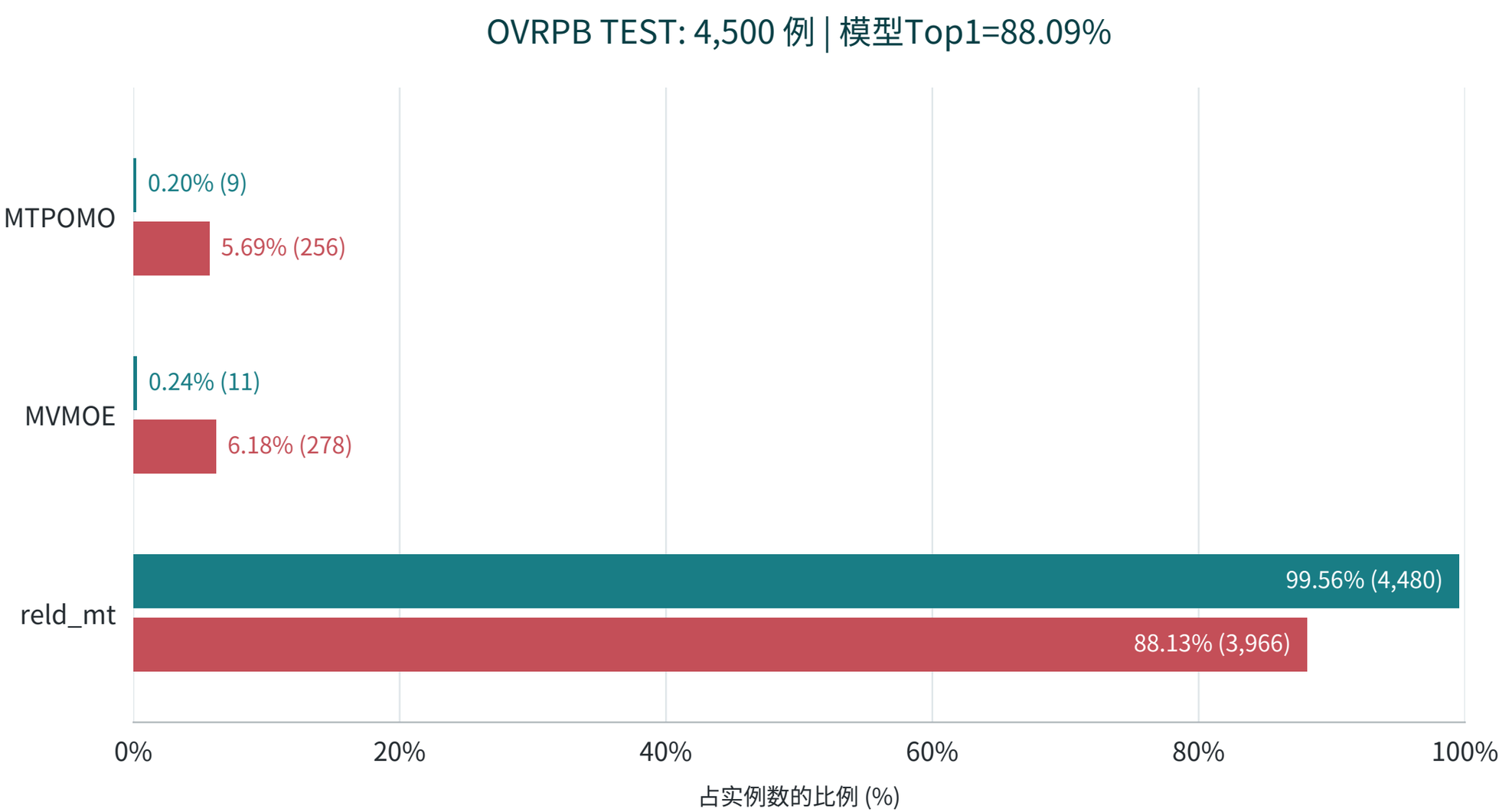
R109/A_tied_late: stable global solver-ID score ties. Same R109 lock; LIB inference only; no training/reselection/Rxx mixing.

R109/A_tied_late | OVRPB | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 2ba5345464b9b022

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

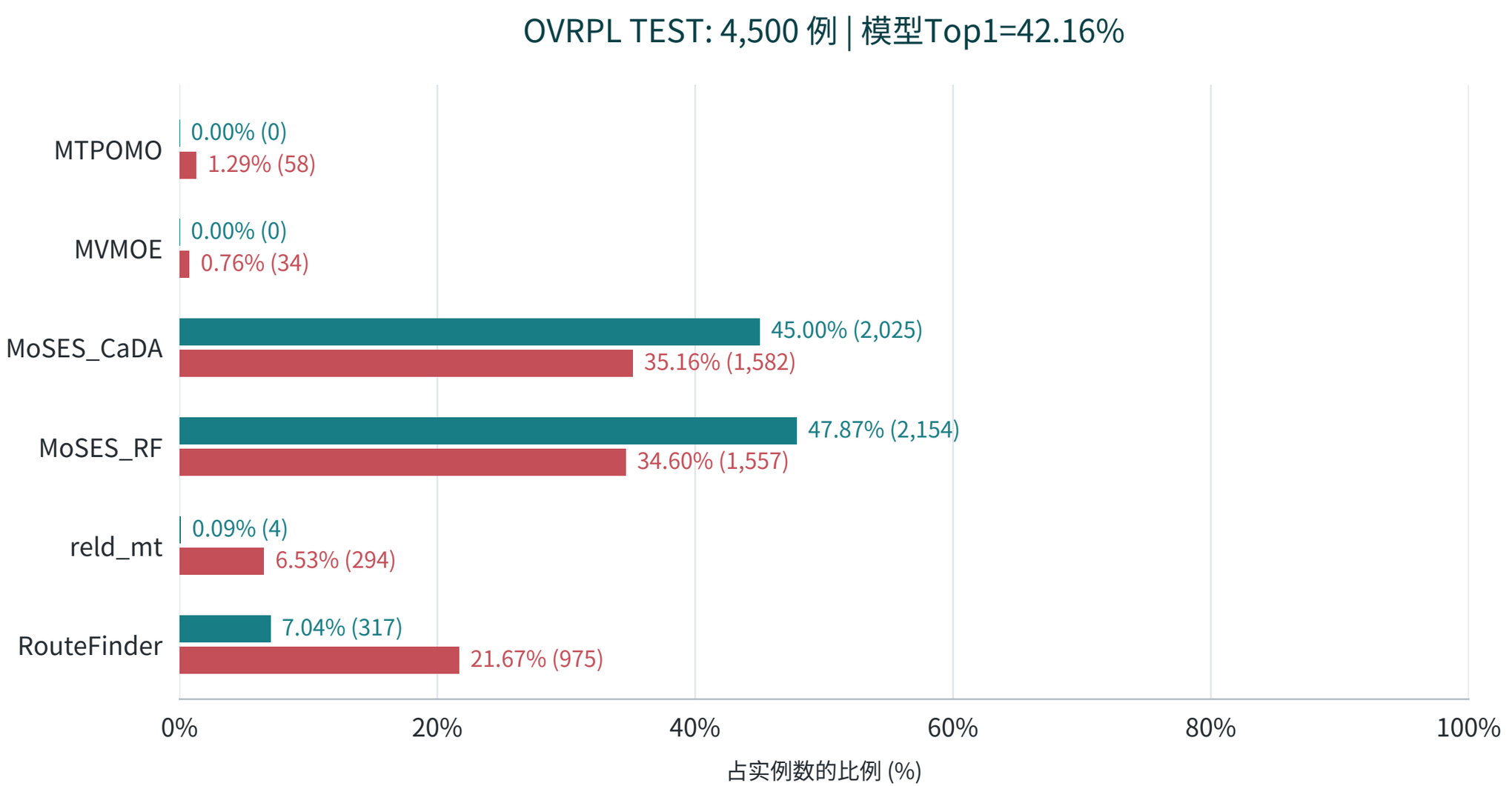
R109/A_tied_late: stable global solver-ID score ties. Same R109 lock; LIB inference only; no training/reselection/Rxx mixing.

R109/A_tied_late | OVRPL | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 2ba5345464b9b022

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

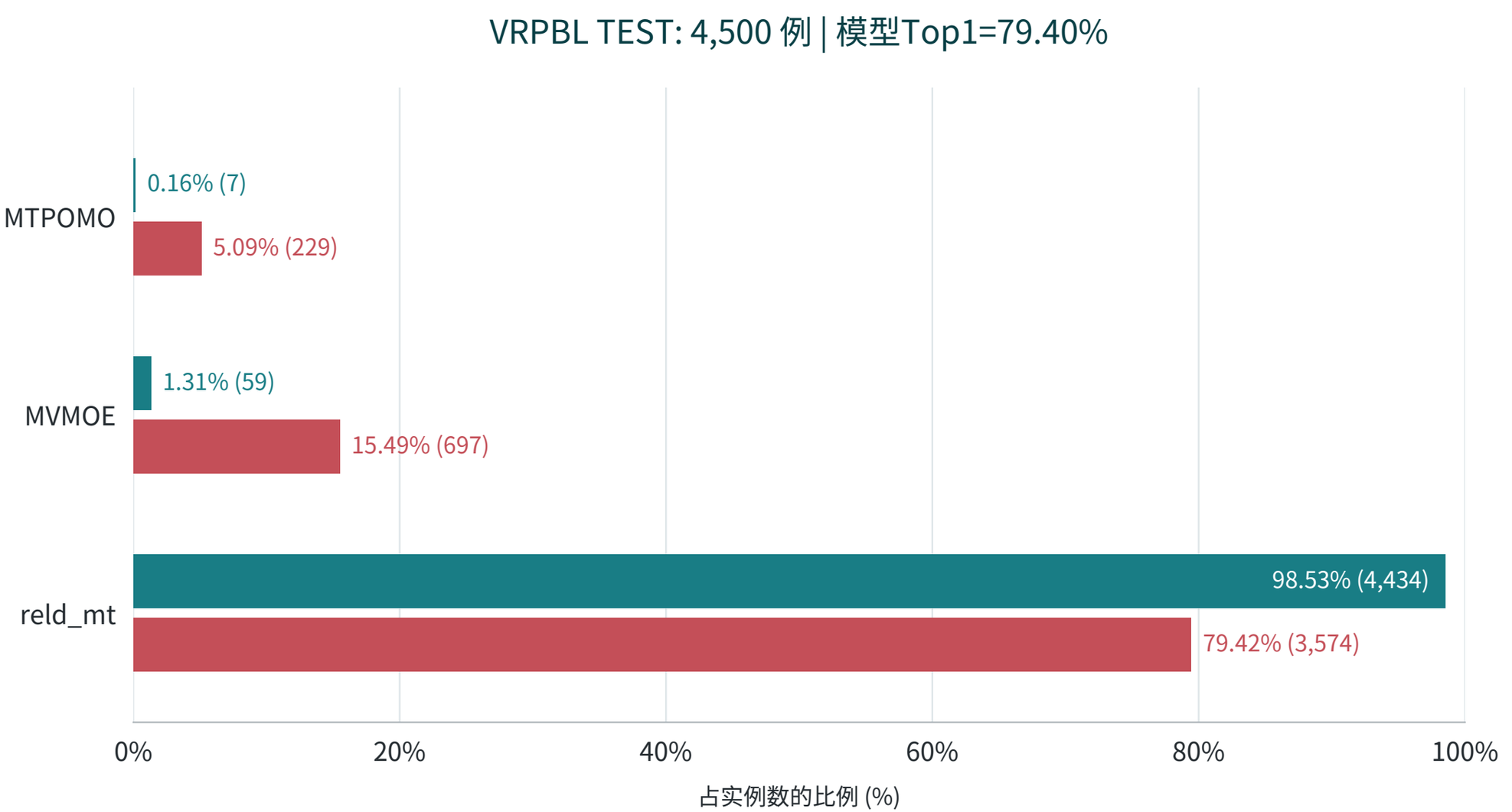
R109/A_tied_late: stable global solver-ID score ties. Same R109 lock; LIB inference only; no training/reselection/Rxx mixing.

R109/A_tied_late | VRPBL | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 2ba5345464b9b022

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

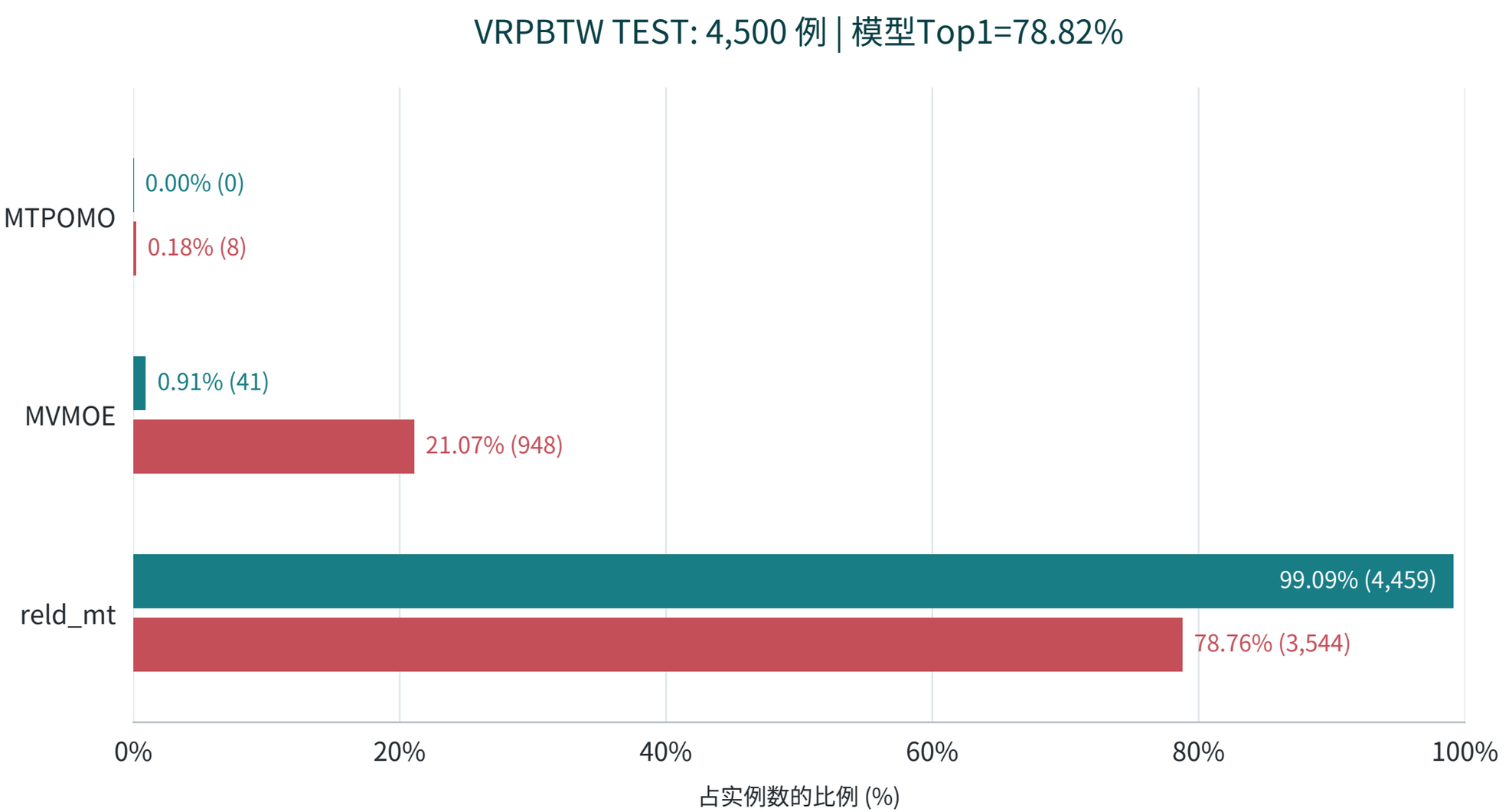
R109/A_tied_late: stable global solver-ID score ties. Same R109 lock; LIB inference only; no training/reselection/Rxx mixing.

R109/A_tied_late | VRPBTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 2ba5345464b9b022

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

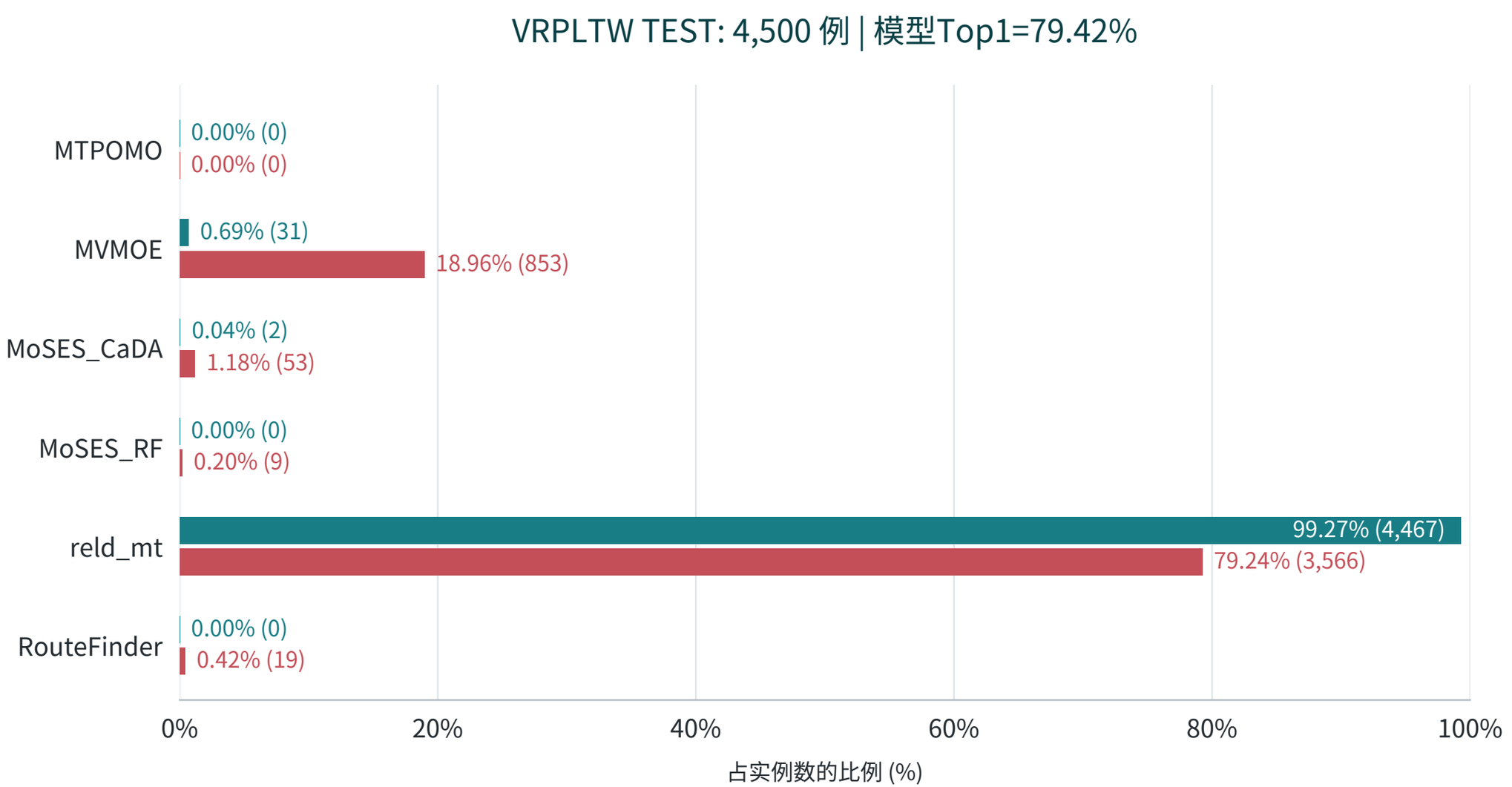
R109/A_tied_late: stable global solver-ID score ties. Same R109 lock; LIB inference only; no training/reselection/Rxx mixing.

R109/A_tied_late | VRPLTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 2ba5345464b9b022

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

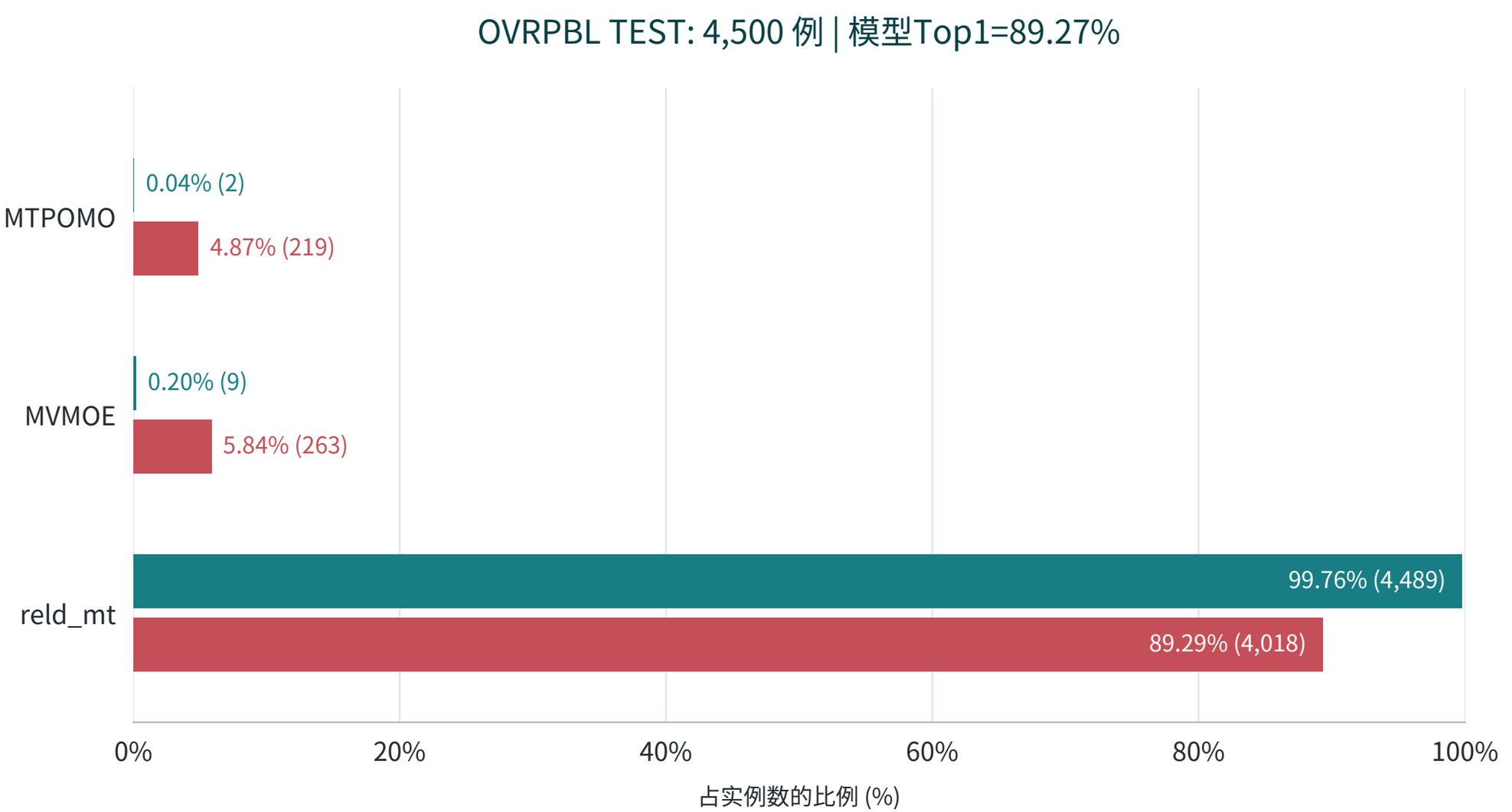
R109/A_tied_late: stable global solver-ID score ties. Same R109 lock; LIB inference only; no training/reselection/Rxx mixing.

R109/A_tied_late | OVRPBL | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 2ba5345464b9b022

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

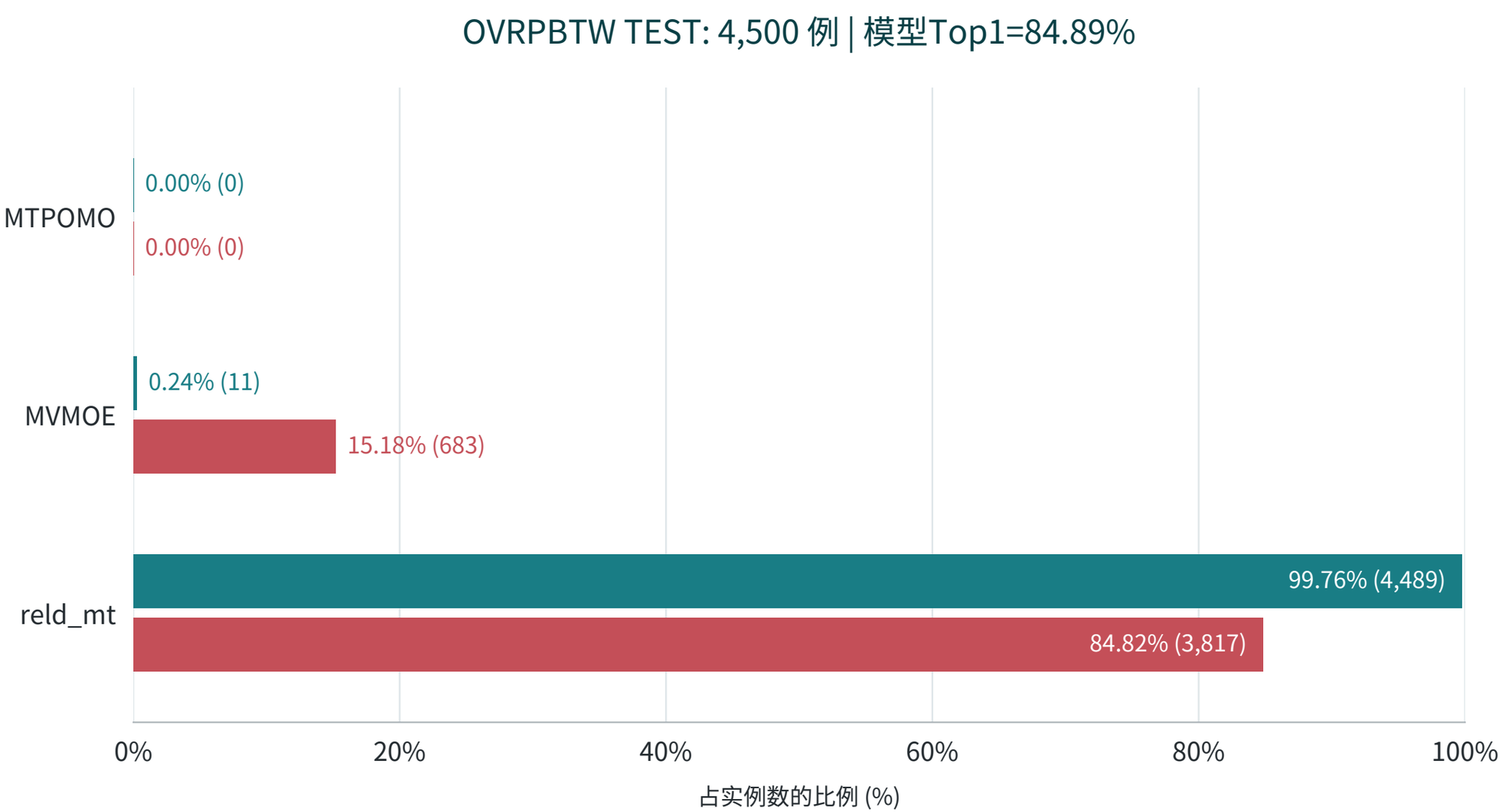
R109/A_tied_late: stable global solver-ID score ties. Same R109 lock; LIB inference only; no training/reselection/Rxx mixing.

R109/A_tied_late | OVRPBTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 2ba5345464b9b022

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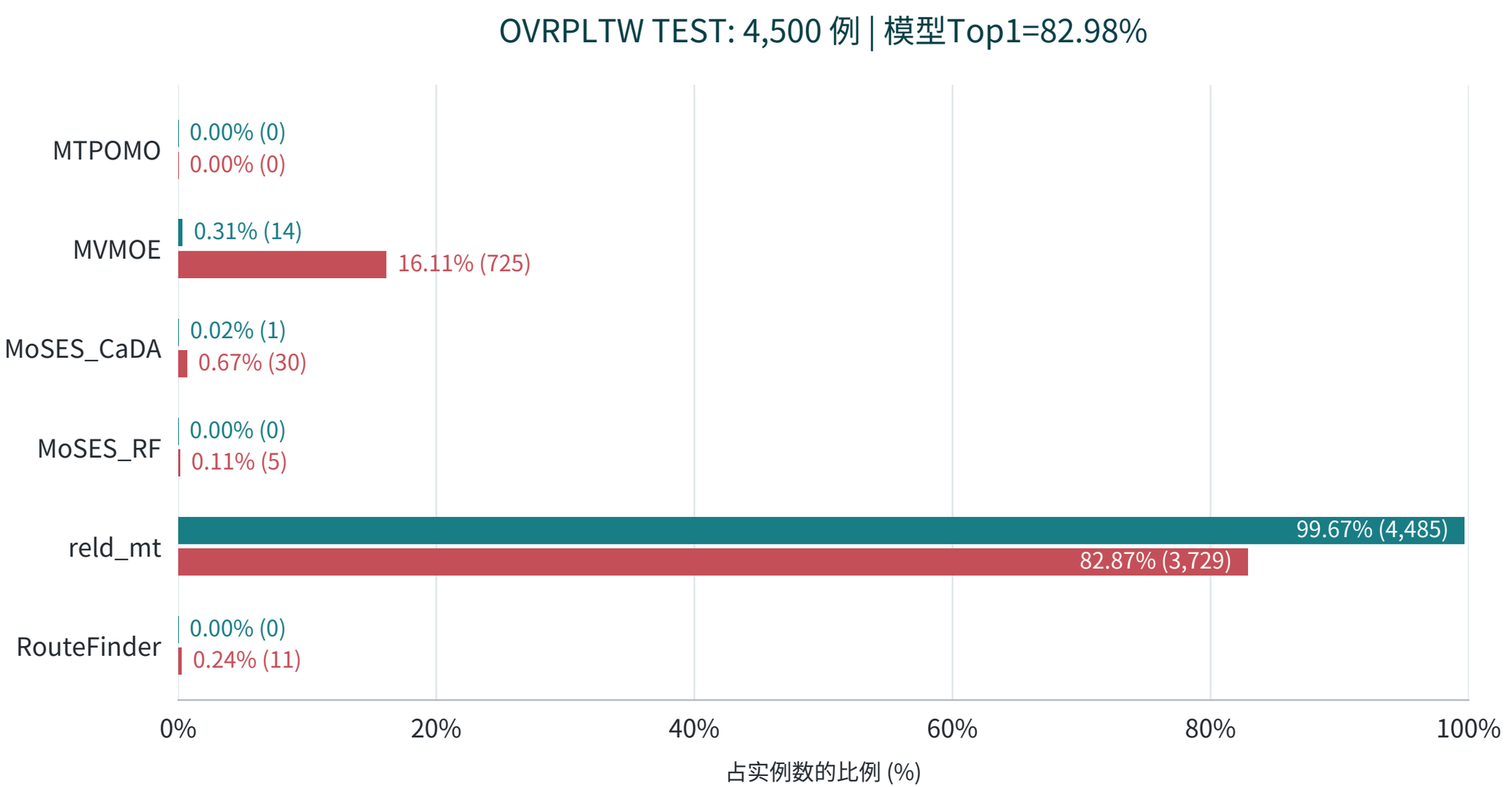
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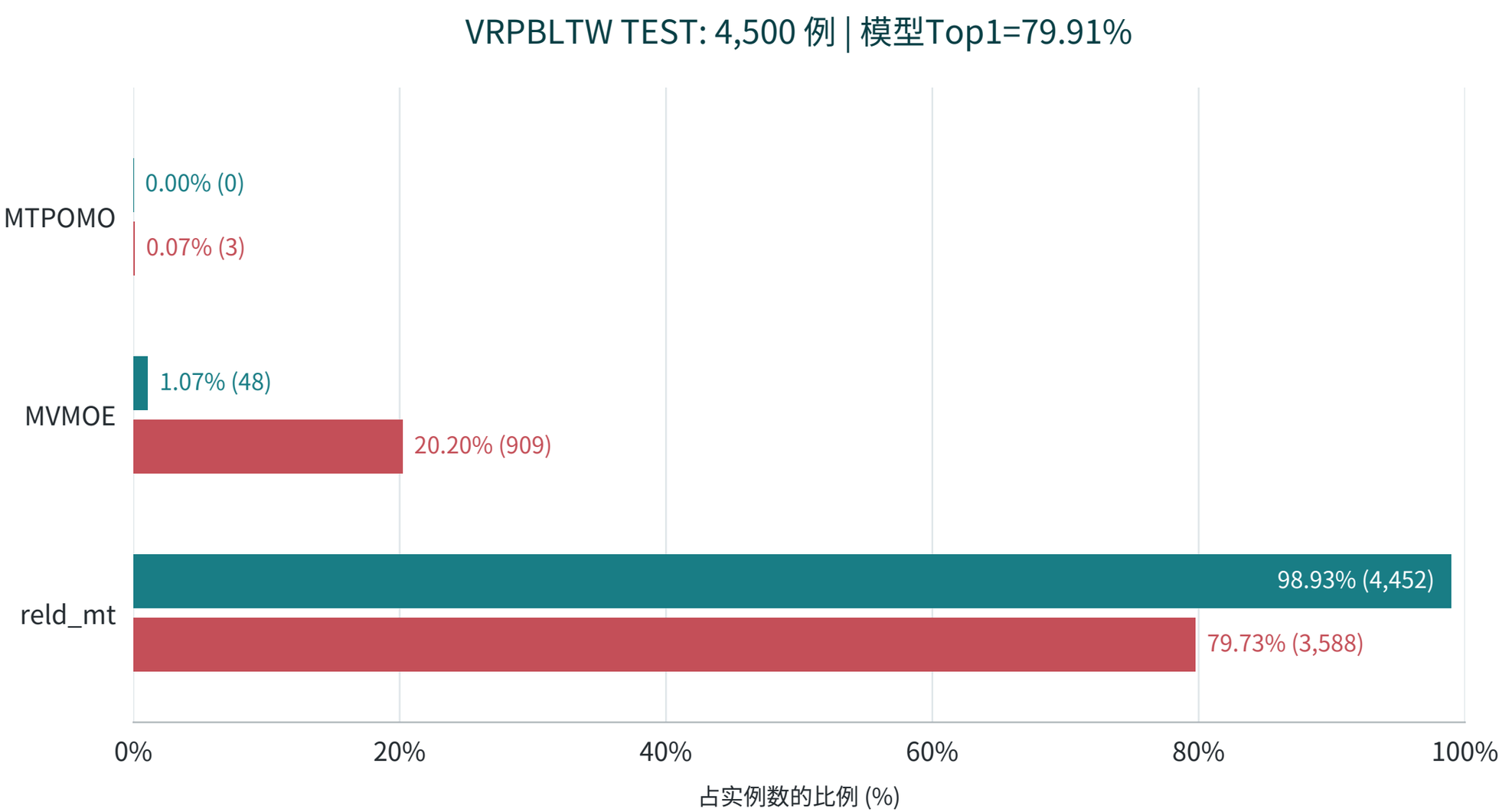
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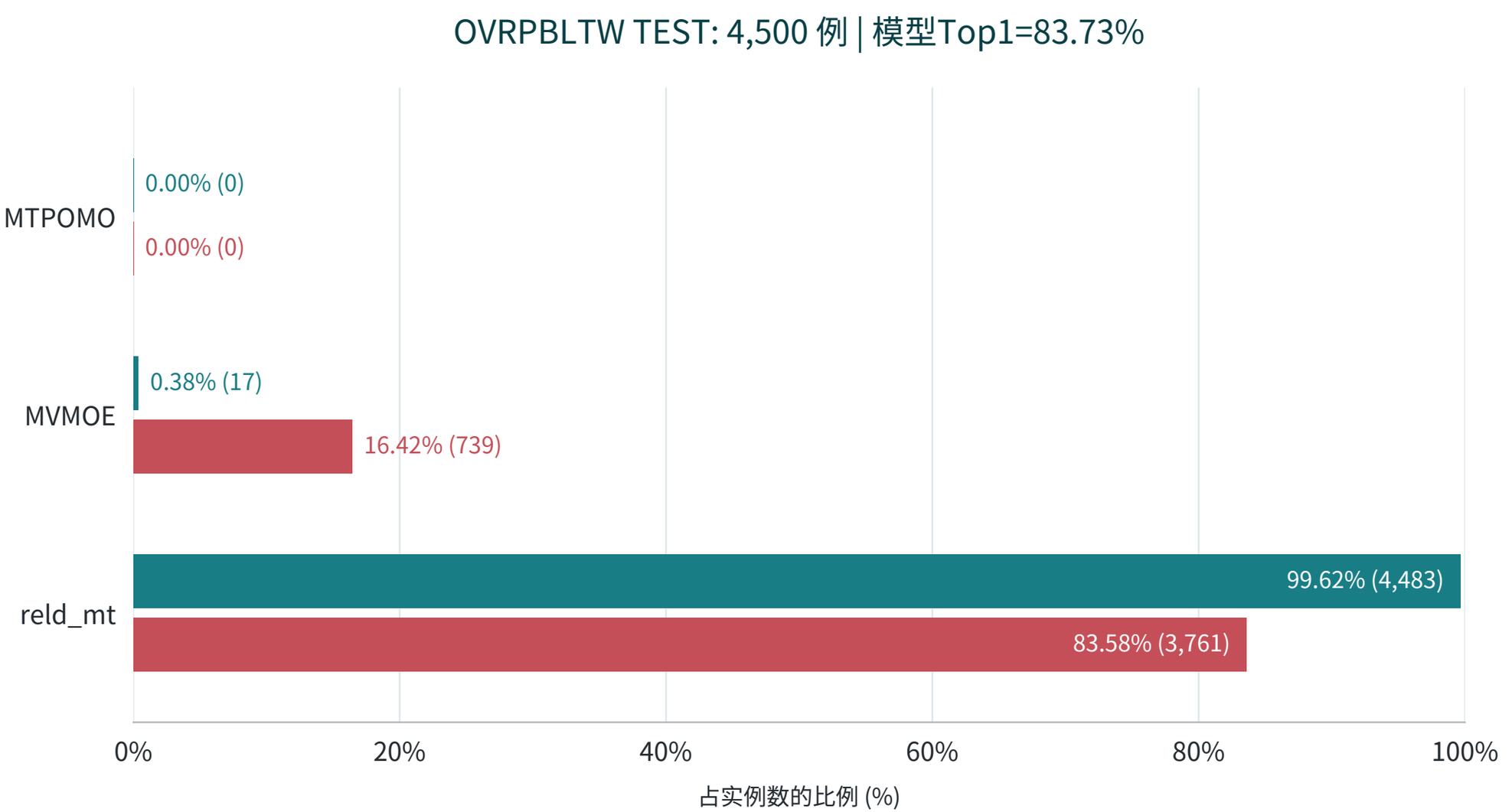
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